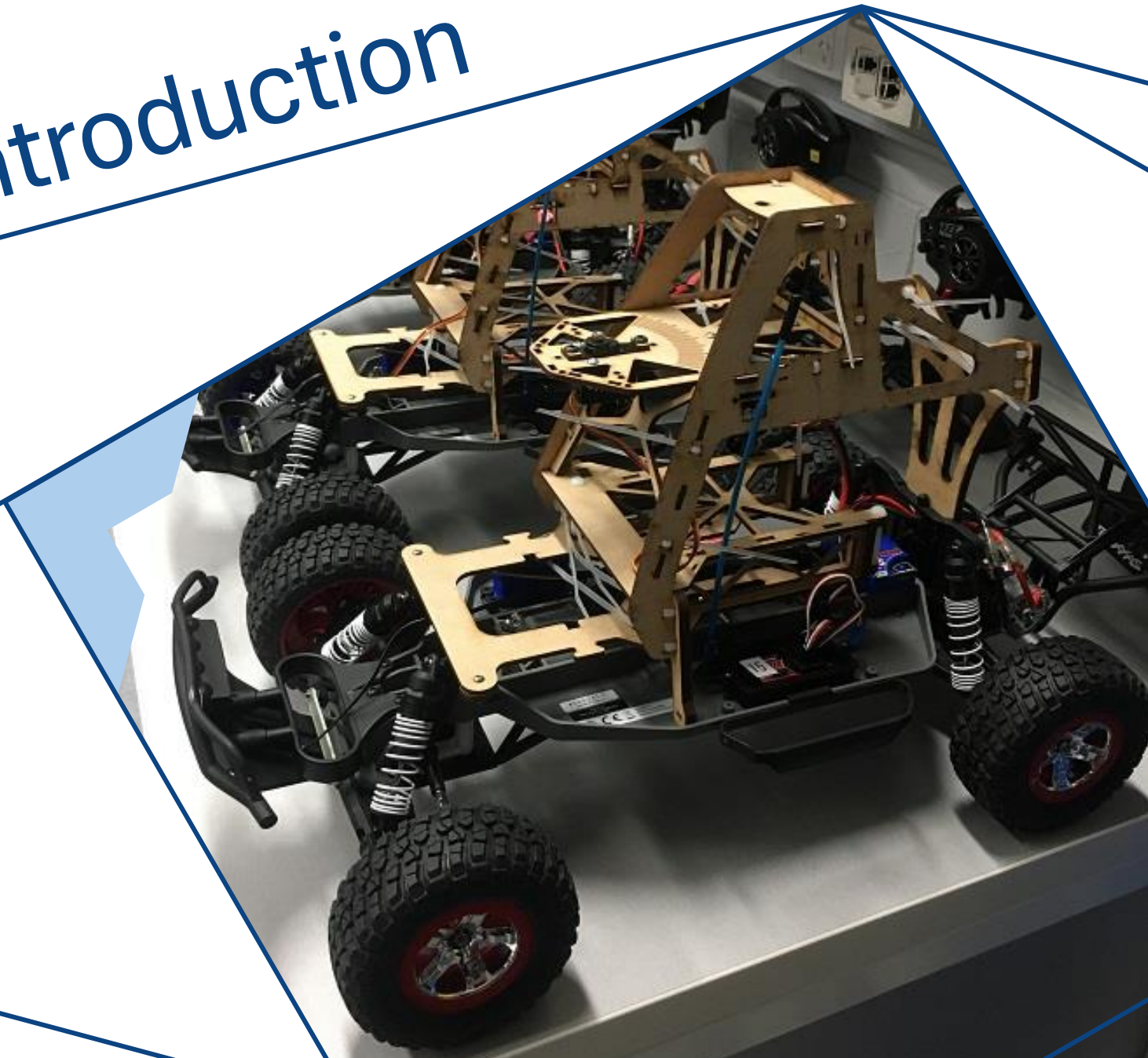


System Project Introduction

Lecture Session

(Rev. 14 Sep 2026)

ELEN90095 AI-4-Robotics



Our real-world hardware platform – “the car”

[OAK-D Pro](#) Stereo Vision Camera

- Onboard chip for running neural network inference.
- We have trained a YoloV8 model for detecting blue and yellow cones.

[RPLidar C1](#) 2D scanning LiDAR for detecting obstacles that are taller than the cones.

Powerbank battery for Jetson and sensors.

Servo for the steering commands.



Inertial Measurement Unit (IMU) ([FSM300](#)) (hidden from view)

[Jetson Orin Nano](#) for running the autonomous system software stack (not present in this photo).

Motor for the drive commands

[Traxxas Slash](#) RC car (2-wheel drive)

A video from 2024 of lane changes

[Link to video on LMS](#)

Team Tiapan

Contents

System Project description and assessment

System Project timeline

Some more details about our hardware platform

How to use the cars (in the Week 8 and 9 Workshop sessions)

Project Description and Assessment

System Project – what and why?

A System Project is one with:

- A **larger team from various disciplines** that are all working towards the same overall goal.
- **Sub-teams work** on sub-systems necessary for the project.
- The whole **team collaboratively integrates** the sub-systems together to achieved their overall goal.

Why in this subject?

- AI4R is rare opportunity with a good mix of different disciplinary backgrounds in one subject.
- Provide experiences that are:
 - 1) Different to the project experience in other subjects.
 - 2) Useful for interviews.



The System Project & interdisciplinary experiences

Exciting AI4R Tech

i.e., learning about architectures & algorithms that make robots do exciting things.

i.e., putting multiple building blocks together into a complete system (PID, RL, MPC, robustness, estimation, automata, ...).

Exciting AI4R Collaborations

i.e., embracing the opportunity to work with and exchange with others on a spectrum of AI4R topics.

i.e., developing and refining what it means for you personally to be an interdisciplinary team player on an AI4R project.

PHASE 1: Engage with AI4R Research (Week 1-3)

Read and present an AI4R paper.

Discuss the paper with peers.

PHASE 2: Individual Upskilling (Week 4-7)

Experience with PID, Deep RL, MPC

Share your experience with peers.

PHASE 3: System Project (Week 8-Exam Period)

Self-defined (and fun!) project

Working with peers as a team.

In the language of interdisciplinarity, this is...

Exploring the boundaries of:

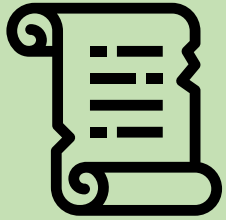
- Knowledge, methods, problem formulations.
- Norms and language of technical details.
- Learning experiences.

In the language of interdisciplinarity, this is...

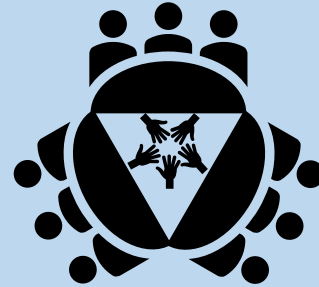
Developing the personal attributes of:

- Listening and entrepreneurship.
- System-level thinking.
- Reflective evaluation.

System Project materials on LMS



Description of the project given on the [System Project: Technical Requirements page](#).



The [System Project: Teamwork Requirements page](#) describes how certain aspects of the team collaboration must be conducted.



Assessment totals 50% of the subject grade and is across 3 main assessments:
[Teamwork Submissions](#) (15%).
[Progress Check-in](#) (0%).
[Final Assessment](#) (35%).

Note: The Task Brief for each assessment will be given directly on the respective assignment pages. (i.e., unlike the presentation and upskilling assignments that had stand-alone Task Brief documents).

Process over product

To mitigate equity challenges with grades for the System Project, the System Project assessment focus on Process and have very little weighting for the product:

PROCESS

Focus on your team demonstrating the process you work through.

PRODUCT

Product is essentially the system behaviour that you are aiming for.
In other words, the product is primarily the motivation

See the FAQ on the System Project: Technical Requirements page:
“Does having more team members mean the project is expected to achieve much better results or include more work?” The answer is one example of what it means for the assessment to focus on process over product.

Theory vs. Implementation

THEORY

We can (usually) craft the theory to tell a clean story.

IMPLEMENTATION

- Implementation (usually) crafts us and the path that we follow.
- Implementation for the System Project needs and benefits from topics that were not part of the lectures.

Connection to lit review and upskilling

Repeating from the previous slide:

Implementation for the System Project needs and benefits from topics that were not part of the lectures.

AI-4-Engineering Systems:

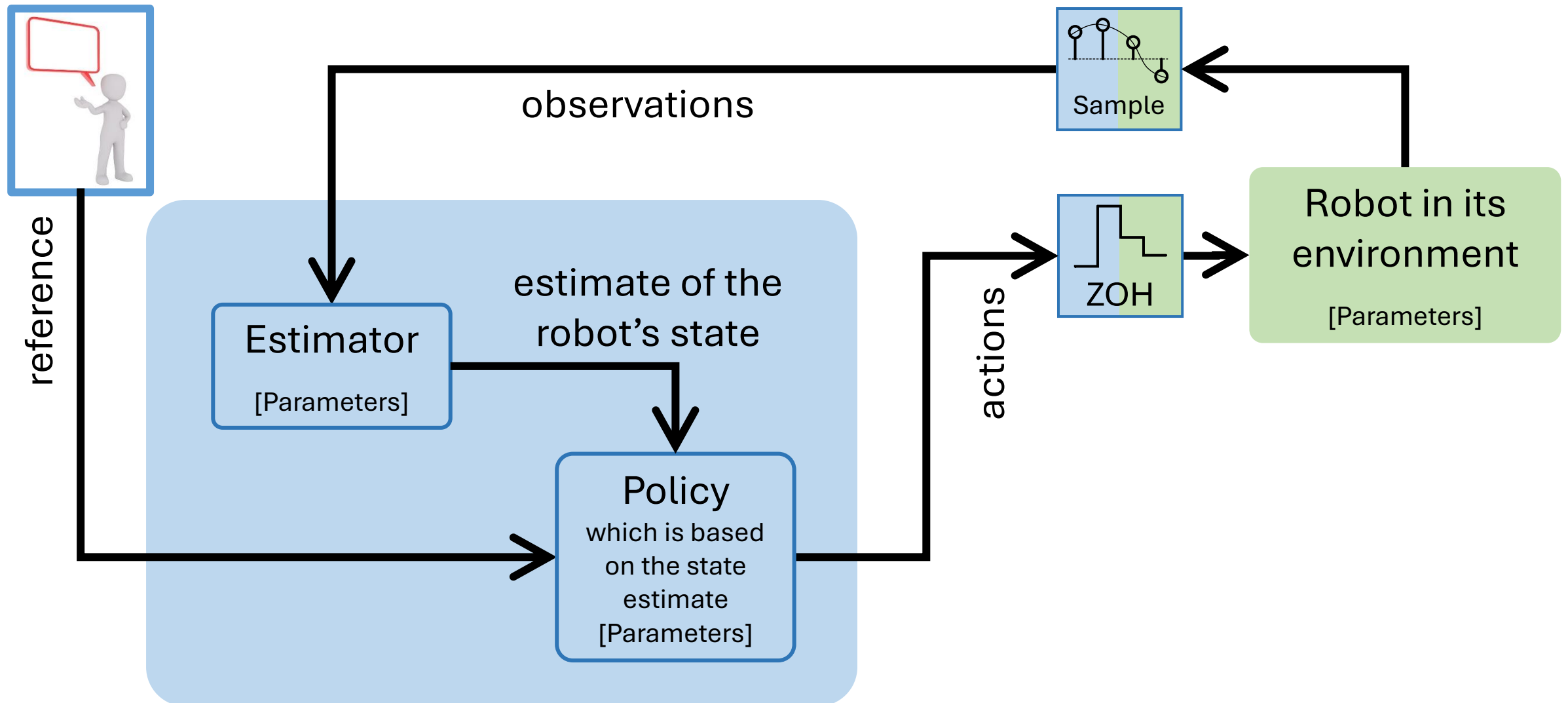
Presentation plus Group Discussion

You can and should search literature for ideas and stimulus of how to design, implement, and test your policies for the System Project

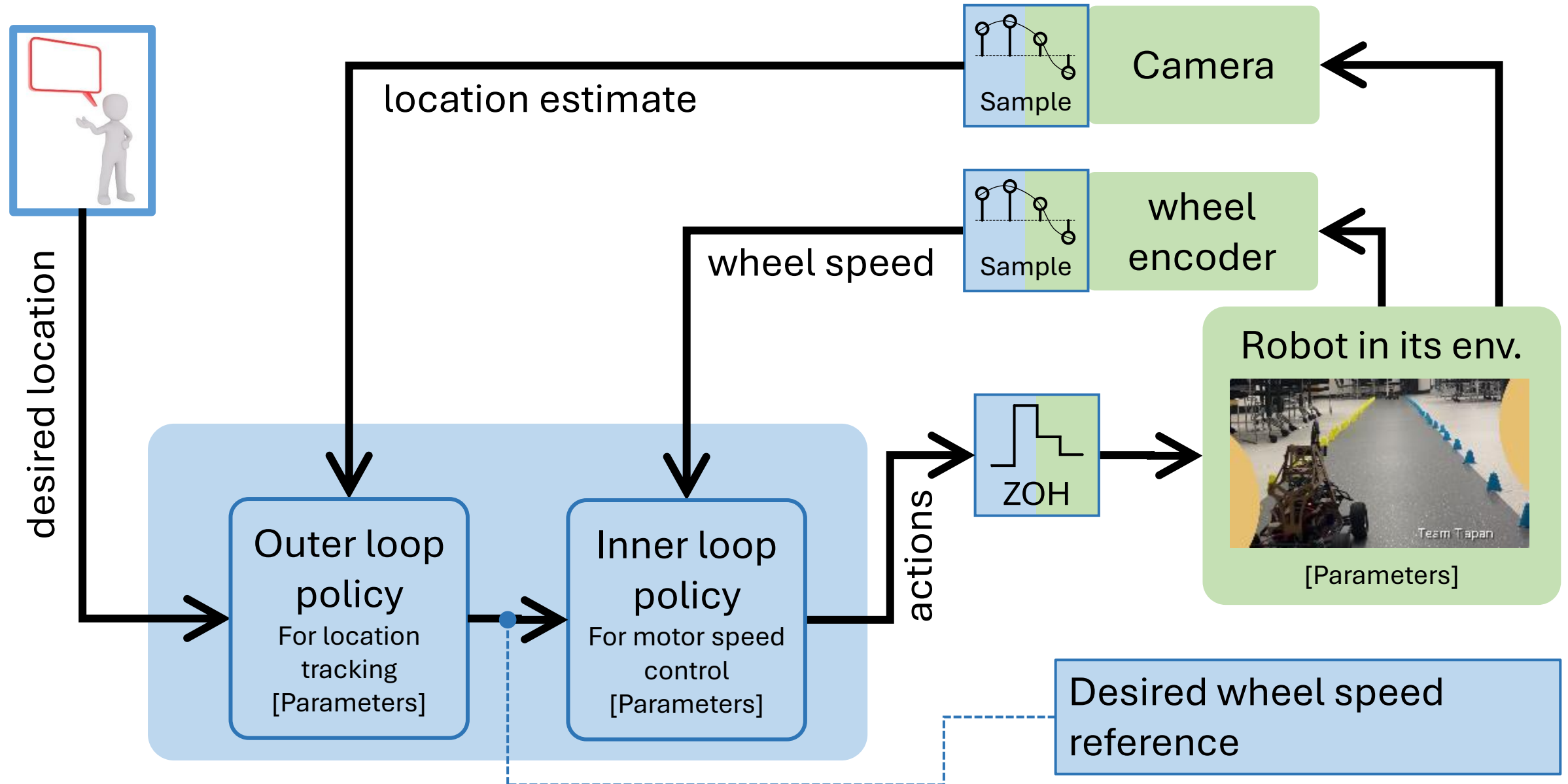
Upskilling

You can and should be using the Gymnasium to simulation your policies, evaluate using performance metrics, test robustness with uncertainty sweeps.

Policy architecture – estimator



Policy architecture – nested loops example



System Project Timeline

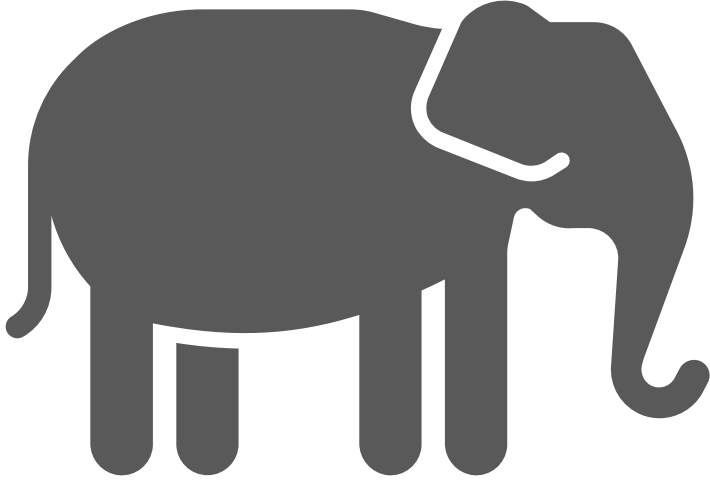
System Project Timeline

Week	Lecture Slot (Mondays 11am-2pm)	Workshop Slot (Tuesdays 9am-3pm)	Assessment
8	This slide deck	Team formation (attendance mandatory)	
9	Technical Content Test	“Hand out” the cars.	
Break	Classroom available.	Classroom available.	
10	Open for consultation.	Project work.	
11	Open for consultation.	Project work.	Progress Check-in
12	Open for consultation.	Project work.	
SwotVac	Classroom available.	Classroom available.	
Exam Period Weeks 1-3	Classroom available.	Classroom available.	Final Assessment

Note: A Teamwork Submission will be required in every week from Week 9 until the exam period (details to come).

Technical Content Test

Details provided on the [Technical Content Test page](#).



In 2025, the test was in Week 11, which we observed to be a significant disruption to the System Project work. Hence **this year the test is in Week 9**.

A reminder of some key percentages:

- **15%** = The value of the Technical Content Test
- **50%** = The cumulative value of the System Project assessments.
- **No hurdle requirement for the subject.**

See respective LMS pages for full details!

IMPORTANT

- This slide deck is **NOT** intended to provide a full description of the System Project.
- The respective LMS pages are the “ground truth”.
- At this point in the lecture, we will switch to LMS and discuss some more details.

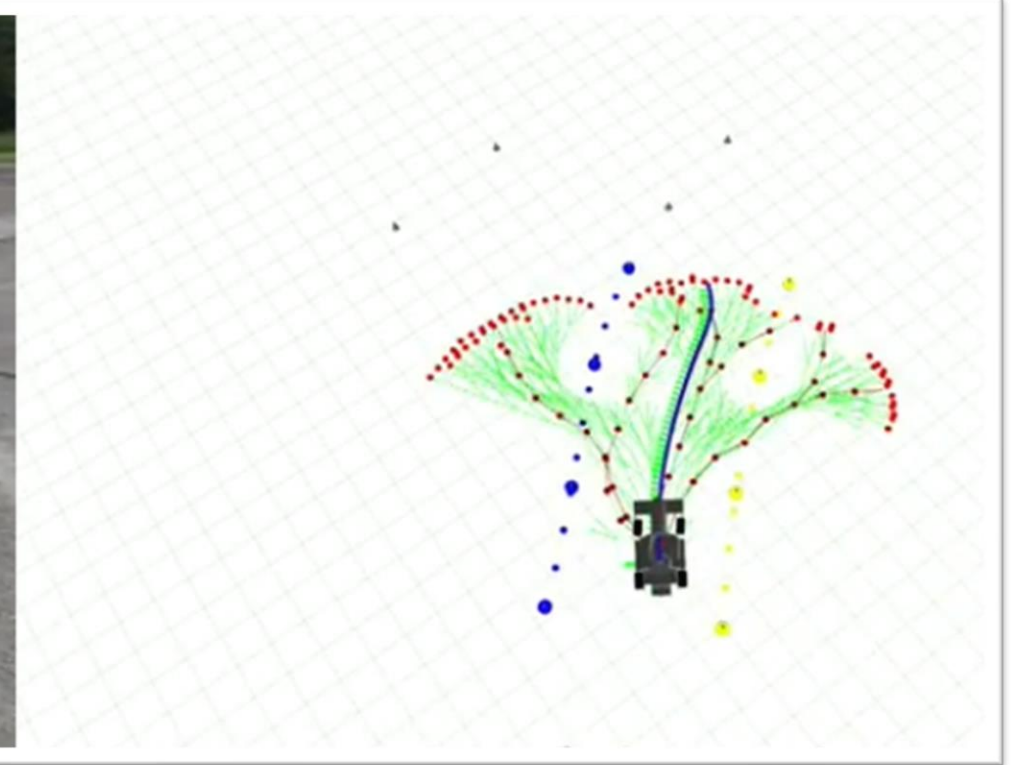
Some more details about our
hardware platform

Are our cars too much of a toy!?



[Link to video on YouTube](#) (and there are many many more humanoid videos out there)

Our cars are inspired by Formula Student



[Link to video on YouTube](#) (and there are more videos if you search “Formula Student Driverless”)

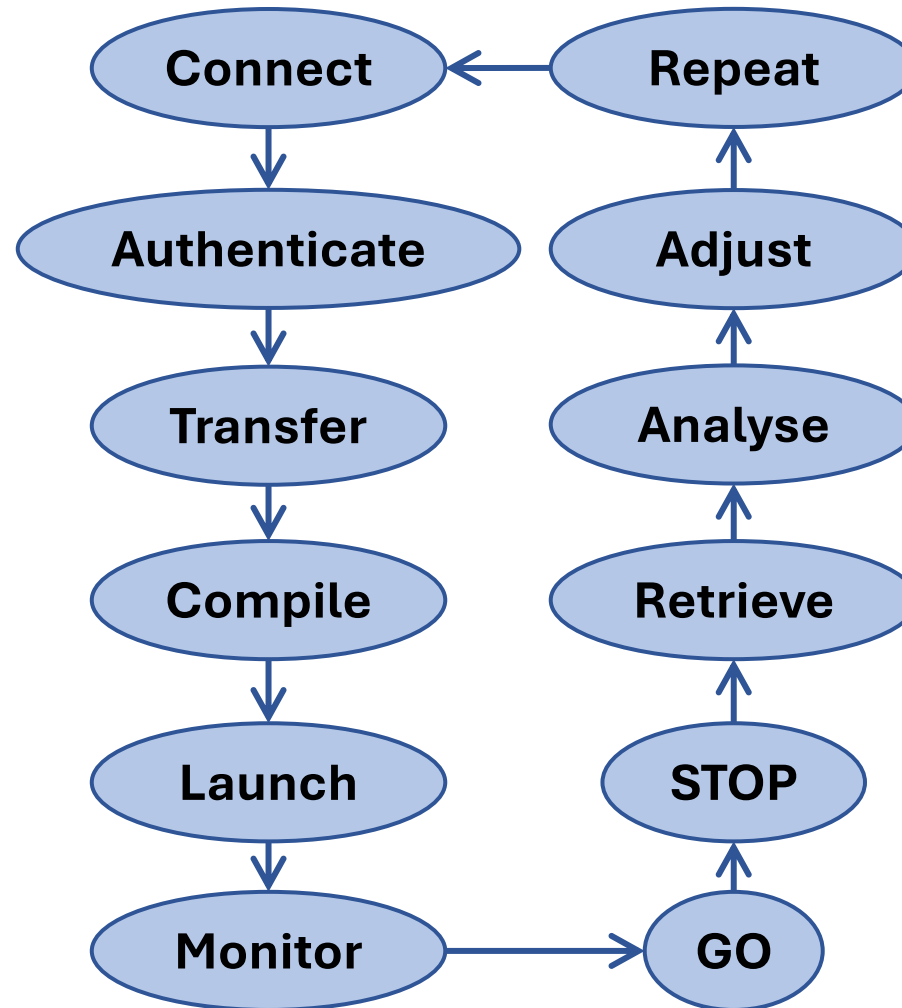
... combine with the scale and popularity of [Robo Racer](#).

What to expect from our cars

Hardware is hard



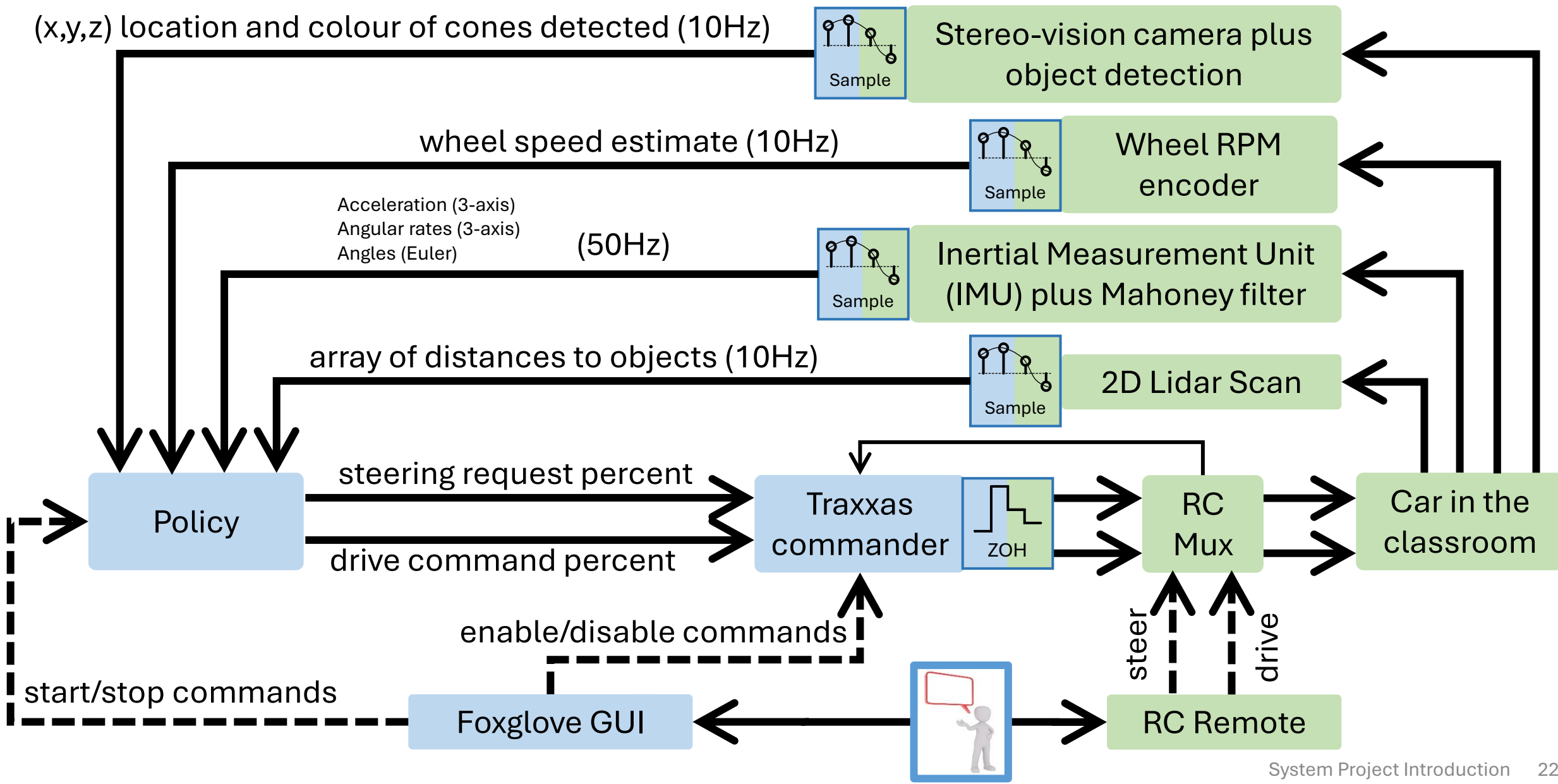
Software is as “easy” as:



Building the ship
as we sail it



Block diagram of the car



“Disclaimer”

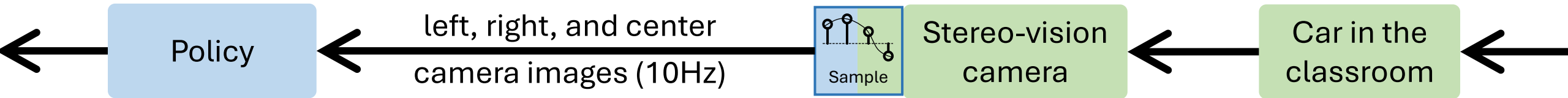
We are running behind with preparing the cars for this year:

- Upgrade camera software to use the latest version of DepthAI.
- Upgrade drive and steer commands to go through a Pico (both human and policy commands).
- Build new bodies that have a place for mounting an RPLidar C1 and that mount the Oak-D camera on a pan servo.
- Upgrade the software so that ~200 people can efficiently share ~17 robots.

Does the following sound believable or does it sound like an excuse?

- Most robotics projects that focus on algorithms spent the first phase on the project in simulation.
- Having the robot available in that first phase provides motivation, even if it has little input to the technical details of the first phase.
- Ask any capstone team who focused on algorithms for robotics.

Camera images as direct input to policy?



- We are NOT doing camera images as direct input to the policy.
- Previous 2 years some students asked for this through the end-of-subject feedback survey.
- I take this as being feedback for a whole new AI4Robotics master's program (and not plausible to fit into this single AI4R subject).
- The Oak-D camera is our friend because it can deploy neural networks for object detection (however it cannot easily deploy general policies).
- For the System Project, a sub-team could choose to work on improving the cone detection pipeline that runs on the Oak-D camera.

Summary

Summary

- Read the relevant pages and assignment on Canvas:
 - [System Project: Technical Requirement page](#).
 - [System Project: Teamwork Requirements page](#).